

Meet Vaghela

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PROFESSIONAL SUMMARY

AI/ML Engineer with 6 months of production experience building LLM agents, RAG systems, and automation workflows. Developed autonomous calling agents that reduced manual task handling by 60% and RAG pipelines improving answer accuracy by 35%. Proficient in LangChain, LangGraph, Semantic Kernel, CrewAI, AutoGen, FastAPI, and C#/.NET. Seeking AI Engineer roles to build scalable, production-ready intelligent systems..

EDUCATION

B.Tech – Computer Science | ITM (SLS) Baroda University

Sept 2022 – Apr 2026

CGPA: 9.20 / 10 · Vadodara, Gujarat

EXPERIENCE

AI/ML Intern | McKH Technologies

Vadodara · Dec 2025 – Apr 2026

- Designed and shipped autonomous AI agents using LangChain + LangGraph, reducing manual task-handling time by an estimated 60% across target workflows.
- Implemented production RAG pipelines (retrieval-augmented generation) integrating vector search (FAISS/Chroma) with LLM inference, improving answer relevance by ~35% over baseline keyword search.
- Automated end-to-end workflows with n8n, Automation/Workflows like Lead-qualified calling agent, Relationship assistant etc.
- Collaborated with the product team to define agent evaluation metrics and iteratively tuned prompts, reducing hallucination rate on domain-specific queries.

Machine Learning Intern | CodeSoft

Remote · 1-Month(2025)

- Built and deployed NLP pipelines (text classification, sentiment analysis) achieving >88% accuracy on benchmark datasets using fine-tuned transformer models.
- Developed a fraud detection model with an XGBoost + neural ensemble, achieving 94% precision on imbalanced financial transaction data.

Machine Learning Intern | CODECRAFT INFOTECH

Remote · 1-Month(2025)

- Designed and trained CNN-based image classification models (ResNet-50, EfficientNet) reaching 91% top-1 accuracy on a custom 10-class dataset.
- Built and benchmarked traditional ML pipelines (SVM, Random Forest) alongside deep learning models, producing a comparative accuracy report that guided architecture selection.

PROJECTS

Apex-Agent (Autonomous Agent | Agentic Assistant) — *React.js · LangChain · LangGraph · FastAPI*

- Architected a Multi-Tiered Sub-Agent (MTSA) framework using LangGraph and FastAPI to orchestrate collaborative AI agents for complex, autonomous problem-solving.
- Integrated the Model Context Protocol (MCP) to securely connect language models with external data sources, file systems, and specialized local tools.
- Developed a highly responsive React.js frontend featuring an Interactive Agent Loop (IAL) for real-time task state visualization and dynamic chat interactions.

GitHub : [Repo Link](#)

Study-Buddy – Multi -Agent AI Learning System— *LangGraph · MCP · A2A · FastAPI · Next.js · CrewAI*

- Designed a multi-agent AI system using LangGraph, MCP, and A2A for curriculum planning, tutoring, quiz generation, and adaptive learning.
- Implemented RAG-based knowledge retrieval with local LLMs (Ollama) and persistent session management using SQLite checkpointing.
- Built and deployed a FastAPI + Next.js application with agent observability (Langfuse) and CrewAI interoperability.

GitHub : [Repo Link](#)

AI-Powered Healthcare Voice Agent & CRM — *Python · Vapi · FastAPI · SQLAlchemy ORM · Next.js*

- Automated Voice Scheduling: Seamlessly handles appointment bookings and cancellations using a low-latency VAPI + FastAPI voice engine with natural language understanding.
- Predictive No-Show Analysis: Features a custom risk-scoring algorithm that identifies "High Risk" appointments based on historical lead times and cancellation trends.
- AI-Driven Admin Intelligence: Provides a Next.js 14 dashboard with real-time analytics, peak-hour visualizations, and automated insights for hospital resource optimization.

GitHub : [Repo Link](#)

Healthcare AI Portal — *Python · ML/DL · LangChain · FastAPI · React.js*

- Built multi-disease prediction models (heart disease, diabetes, lung cancer) using fine-tuned ML classifiers, reaching 87–92% accuracy across all three domains.
- Developed an interactive medical chatbot (LangChain + RAG) for personalised risk assessment, reducing time-to-guidance compared to static decision trees.

GitHub : [Repo Link](#)

N8N Automation — *LLM, VAPI, N8N, AI Tools*

- Built an autonomous lead qualification calling agent using *n8n* + *Vapi*, capable of initiating calls, interacting with users, and updating CRM(sheet) workflows without human intervention.
- Designed end-to-end agentic workflow automation, including lead filtering, conversation handling, and follow-up scheduling, improving response efficiency.
- Developed a domain-specific AI relationship assistant leveraging structured workflows and LLM orchestration to deliver contextual, human-like interactions.
- Engineered RAG-based and RAG 2.0 systems integrating vector databases and LLMs to deliver accurate, context-aware responses from custom knowledge bases.

TECHNICAL SKILLS

AI / LLM Frameworks: LangChain, LangGraph, RAG, LLM Agents, Prompt Engineering, FAISS, ChromaDB, Crew AI, AutoGen, Semantic Kernel

Languages: Python, SQL, JavaScript, C#

Backend & APIs: FastAPI, Flask, REST APIs

Frontend: React.js

Databases: MySQL, MongoDB

DevOps & Tools: Git, Docker, n8n, vapi

CERTIFICATIONS

- Gen-AI : IBM
- Python – HackerRank
- AI Engineer – OneRoadmap
- Data Analysis & Visualisation – IBM / Google / Microsoft
- LangChain for LLM Application Development – CodeSignal
- AI – Anthropic